Cilium Open-Source Dev Summit Part 2

Mar 23, 2026

Summary

Data path plugins were approved for eBPF extension, while high-scale identity allocation and IP cache limits were identified as critical scaling bottlenecks.

Data Path Plugins Approved
The Data Path Plugins CFP has been approved, and the Pull Request is available for review, enabling additional eBPF program loading via a gRPC interface. Next steps include improving ewe coverage and addressing current limitations in tail call instrumentation.

Scaling Bottlenecks Identified
The identity allocator is a major bottleneck due to the 65536 identity limit, driven by deployment tools adding labels, which is insufficient for 300000-pod targets. The IP cache limit of 512k was also noted as a scaling concern for dual-stack scenarios, prompting an issue to investigate its configurability.

Testing Tool Proposed
A tool tentatively named 'kube rewind' was proposed to record and replay resource state changes from expensive scale tests in a smaller, localized environment like Kind. This approach aims to provide faster feedback loops, making high-scale debugging and performance metric generation more maintainable and accessible.

Next steps

- [ ] Someone in Baluty (pl-waw-rodc2) will share more details on the idea of using LPM try instead of Arena shared map for the policy engine, possibly at the next week's meeting.
- [ ] Someone in Hoge Veluwe (nl-ams-cla) will file an issue regarding the IP cache limit of 512k and look for context on why that limit is in place.
- [ ] Someone in Hoge Veluwe (nl-ams-cla) will look for better E2E coverage of the plugins.
- [ ] The group will take a look at the data path plugins to see if it can help with inserting tail calls and jumping from them.
- [ ] Someone in Hoge Veluwe (nl-ams-cla) will try to come up with better ways to bring a better tooling experience for scale tests.
- [ ] Someone in Hoge Veluwe (nl-ams-cla) will be taking notes on input regarding the list of items to discuss for potential retirement or graduation.

Details

- **Data Path Plugins Update and Next Steps**: The data path plugins CFP has been approved, and the Pull Request (PR) is available. Data path plugins enable loading additional programs to interact with eBPF, and a gRPC interface has been created by Jordan to manage these plugins (00:00:00). Next steps include improving ewe coverage, addressing limitations in tail call instrumentation, and exploring future enhancements such as plug-in instrumented efilers or k-funs (00:01:14).
- **Data Path Plugin Configuration**: someone in Hoge Veluwe (nl-ams-cla) noted that Jordan is exploring the use of DPG gen to generate proto IDL, which they suggest is intended to help address an issue with passing constants to the plugin (00:02:27). A demo is available that shows the deployment and uninstallation of a data path plugin, including tracing. Someone in Hoge Veluwe (nl-ams-cla) suggests reviewing the available demo (00:03:48).

- **Scaling Bottlenecks: Identity Allocator**: someone in Baluty (pl-waw-rodc2) identified the identity allocator as a significant bottleneck in the current eBPF implementation, noting that customers have experienced exhaustion of identities even at lower scales (00:03:48). The core issue is that customers may not fully understand what labels and identities are, sometimes exhausting the 64k identity limit due to specific deployment tools that add labels. Someone in Baluty (pl-waw-rodc2) emphasized the need to address this limitation to support high scale targets like 300k pods (00:05:30) (00:11:20).
- **Identity Allocation: User Experience and Configuration**: someone in Hoge Veluwe (nl-ams-cla) acknowledged that while guiding users to limit relevant labels could work for those with deep knowledge, it is not user-friendly for most (00:05:30). They suggested improving the user experience, potentially by advocating for upstream changes like tying identity directly to the service account namespace, which could eliminate many issues (00:06:57). The current design allows any label to become relevant, which complicates the identity mechanism (00:08:01).
- **Identity Space Utilization and Cluster Mesh**: someone in Baluty (pl-waw-rodc2) pointed out that in the current implementation, the cluster mesh part of the identity space can "steal" bits from the identity space, but not the other way around (00:09:19). They proposed allowing users who do not utilize cluster mesh to use the bits allocated for cluster mesh. Someone in Hoge Veluwe (nl-ams-cla) expressed concern that such a change could cause conflicts if users later decide to adopt cluster mesh (00:11:20) (00:17:11).
- **Policy-Based Dynamic Label Allocation**: someone in Hoge Veluwe (nl-ams-cla) recalled a previous discussion about dynamically allocating security-relevant labels by inspecting the network policy object. Although this approach focuses on the necessary labels, it results in high churn whenever network policies change, creating rollout issues (00:12:44). Someone in Hoge Veluwe (nl-ams-cla) suggested finding a middle ground, such as a slow-roll migration process, to reduce churn while using more identity space (00:14:04).
- **IP Cache Limits and Configurability**: someone in Baluty (pl-waw-rodc2) noted that the IP cache is capped at 512k, which could be insufficient for scenarios requiring dual-stack support with 300k pods, an issue that will become more acute as Google scales. Someone in Hoge Veluwe (nl-ams-cla) indicated that this setting is likely configurable, suggesting that the problem might be a memory limitation (00:19:44). The participants agreed to file an issue to investigate if this limit is due to kernel restrictions or if it is just a configurable number (00:20:59).

- **Scaling Bottlenecks: Mutex and Endpoint Regeneration**: someone in Baluty (pl-waw-rodc2) identified a mutex implementation in the policy engine that causes performance degradation at very high scales (00:22:08). They also noted that endpoint regeneration takes a long time for some customers who have high pod churn (00:23:38) (00:28:17). Someone in Hoge Veluwe (nl-ams-cla) suggested demonstrating this performance issue with a unit test level benchmark, as this would be necessary to validate any proposed fix (00:24:54).
- **Compilation Lock Discussion**: someone in Hoge Veluwe (nl-ams-cla) noted that the kernel has a single global lock for loading BPF programs, which is likely where the majority of time is spent, regardless of optimizations made in eBPF. They suggested that if compilation is running beyond the initial startup, which should not happen if parameterization is used, the relevant data should be submitted for analysis (00:27:03). Someone in Hoge Veluwe (nl-ams-cla) offered to provide more context on the regeneration time metric and suggested creating a reproduction test case to understand the customer use case (00:29:21).
- **L7 Proxy and Arena Shared Map**: someone in Baluty (pl-waw-rodc2) noted that the L7 proxy function `creator update redirect` is called for every filter and can consume significant CPU time at scale (00:30:44). Someone in Baluty (pl-waw-rodc2) also proposed using an arena shared map in the policy engine to maximize space saving through duplication. However, due to complications with arena support and its requirement for kernel version 6.9+, it was determined to be a future consideration, and they are now considering using an LPM trie (00:33:26).
- **Arena Implementation Complexity**: someone in Baluty (pl-waw-rodc2) and someone in Hoge Veluwe (nl-ams-cla) agreed that while the arena shared map is a powerful idea for space saving, its current implementation is difficult, as documentation is sparse and the verifier throws errors when trying to cast an address to an arena pointer (00:34:54). Someone in Hoge Veluwe (nl-ams-cla) suggested that using the arena might be better for an experimental feature, not the policy engine, until better knowledge is developed (00:38:07). Someone in Baluty (pl-waw-rodc2) noted that using an LPM trie might be unavoidable due to the need to support port ranges (00:39:42).
- **Tracing and Scalability Testing Challenges**: someone in Hoge Veluwe (nl-ams-cla) noted that while tracing was previously discussed, work is needed to complete the flowjob, which involves generating packets and subsequent handling (00:45:58). The discussion then shifted to the challenge of scalability testing, where current high-scale tests are massively expensive, have terrible

feedback loops, and are difficult to maintain (00:47:35). Someone in Hoge Veluwe (nl-ams-cla) proposed that a new approach is needed to better support high-scale targets (00:48:55).

- **Proposal for State Recording and Replay Tool**: someone in Hoge Veluwe (nl-ams-cla) introduced a concept for a tool, tentatively named "kube rewind," to record the state and changes of specified resources (nodes, endpoints, etc.) during expensive scale tests. This tool would generate a test artifact that could be shared with the community, allowing them to reconstruct and replay the complex state of a high-scale test in a much smaller, localized environment, such as Kind (00:50:07). The goal is to simplify test creation and enable developers to reproduce issues without needing massive resources (00:53:12).
- **Tool for Debugging and Scale Testing**: someone in Hoge Veluwe (nl-ams-cla) introduced a tool designed to simplify scale testing and debugging for Kubernetes resources, acting as a `cubectl` extension that adds time information to resource queries. The tool operates by deploying an operator that watches the Kubernetes API for changes to chosen resources, saving the data to storage like Google Cloud Storage or AWS S3. This approach aims to address the complexity of current tooling and the difficulty in analyzing millions of log entries, making it easier to track changes and identify failures at specific times (00:59:28).
- **Benefits for Scale Testing and Metrics**: The tool is expected to improve the efficiency of scale testing, particularly for large-scale environments like 15,000-node tests, by allowing for the recording of tests and subsequent parallel runs with different instances without re-running the full scale test repeatedly. This capability could potentially make it possible to generate performance metrics for different instance sizes at high scale, which is currently considered impossible (01:01:56). The utility of the tool is being tested against known, reproducible scenarios, such as slow startup limits related to Cilium nodes in GKE and issues with endpoint slices (01:03:26).
- **Addressing Debugging Challenges**: The new tool seeks to resolve the challenge of debugging scale issues, which often requires adding deep logs and re-deploying in development environments, a process that is time-consuming and difficult, especially when running software like Cilium in debug mode at high node counts (01:04:54). The goal is to provide faster feedback loops and make debugging more maintainable and democratic across the open source community (01:07:12). The current process often catches regressions very late, usually after a version has already been deployed to GKE (01:08:16).

- **Data Sharing and Anonymization**: The discussion covered the complexities of integrating user-submitted issues and data into the system, particularly concerning security and privacy, which would necessitate anonymization logic for shared information like service names (01:08:16). The current best practice for sharing system snapshots involves providing users with a tool to capture data, but leaving the responsibility of anonymizing and determining what to share entirely up to the user to avoid liability issues. The participants acknowledged that the tool, at least initially, is likely intended for testing purposes and might not be safe to run in production environments without further security consideration and optimization (01:09:34).
- **Review of Project Features and Maintenance Burden**: The conversation shifted to a review of existing project features to assess which ones should be prioritized for improvement and which ones might need to be retired to reduce the overall maintenance burden (01:13:29). Categories for review include the feature's name, its area in the codebase, its project goal, community usage, and the maintenance load it imposes. Endpoint Slices was highlighted as a very significant feature that is widely used, particularly in GKE and Azure (01:15:21).
- **Status of BPFT Datapath Implementation**: The status of the BPFT datapath implementation was discussed, noting that its development stalled due to the lack of time to chase down all related bugs and limited user demand (01:17:33). The feature aims to move packet processing out of IP tables and into BPFT for improved performance, though it currently has issues with certain network configurations, such as incompatibility with Netkit, which may require kernel patching (01:18:42). The consensus was that while it is an interesting point, the current small code path is not causing significant issues, making it a lower priority for immediate attention (01:20:54).
- **Community Input and Meeting Scheduling**: It was suggested that the review of project features should be conducted within a community meeting to gather broader input. The community meeting is typically held on Wednesdays, with the time usually falling around 4 or 5 p.m. for those in the US and Europe (01:20:54). Netkit was discussed as a feature that aligns with the project’s goal, primarily for performance, but the timeline for its full maturity and adoption remains uncertain (01:22:37).

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00:00:00

Hoge Veluwe (nl-ams-cla): All right. Okay. So, we're going to push on through. This is the afternoon session and um slideshow. So, uh something is here on the call. So, we're going to go over a couple of topics. I'm going to be presenting stuff from Jordan and then will also be presenting. So the first thing I just want to discuss um kind of not in huge detail although we can get into it if we want is the data path plugins. So hooray the CP has been approved and I think the PR is up. So just a brief recap. Um basically data path plugins is a way to load additional programs to interact with the selium bpf uh basically in front of and behind. And essentially what Jordan has done is created this like gRPC interface to kind of um interact with the various plugins. And um I think if you want more details, it's all up in the CFP, but the kind of bottom line is that the plugins uh CF merged and then the PR is up.

00:01:14

Hoge Veluwe (nl-ams-cla): So apparently Jordan has been seems he's been working on it for the past few months and I think it's like ready to people kick the tires. So I think um in terms of next steps so definitely we need better ewe coverage um of the plugins and then I think he's hit some limitations on tail call instrumentation so he's been trying to find some alternatives there and then he's looking at potentially even future enhancements which is like can you have a plug-in instrument efelers or k funs and then also in terms of like map replacements like uh yeah telling selium how to deal with the data structures not just the code. So um I don't know have you guys taken a look at that at all. It's kind of a cool thing because it Yeah, it is because it might help us because we we are actually working selium or you know uh inserting our own tail calls jumping from uh instead of like using standard tail call what does we jump our part and if we decide to we jump back to where we initially like it out.

00:02:27

Hoge Veluwe (nl-ams-cla): It could help us a lot as well. We should take a look at that. Yeah, please like I I think this is the time to give like tolerance of feedback like hey is this working not working? Um yeah like shaving it right now. I think the other thing in terms of next steps was like I think we were looking at like if you eliminated tail calls with the BPF to BPF function calls instead there could be easy ways to like replace function that's like very future. Yes. So so the the shell call has to happen before that. Um any comments on that? Oh okay. So there's this other thing. So, um I think there is an issue with sort of passing kind of constants to the plugin, but there's no what Jordan is looking at is using DPG gen and if he's he's getting some things that needs to be worked out essentially uh using the DPGEN to generate photo IDL that then goss and uh anyways this this will be a discussion I think on his uh PR to see what to do here.

00:03:48

Hoge Veluwe (nl-ams-cla): All right. Oh, there's like a demo play here. Just also type in password. So, I think um this demo uh basically deploys a data path plugin and then uninstalls it as well. demonstrates installing it and then having all the traces come out and then anyways I'd say give it a look back. All right scaling so want to talk through these
Baluty (pl-waw-rodc2): Sure. Um so I take a look how uh current selium implements the policy engine and um this is the bottlenecks I see um with the current implementation. I think biggest one is that identity allocator. So in Google so even the very lower scale we see one of customers had the problems that uh they exhausted the um identities and I see here a few problems first of all they
Hoge Veluwe (nl-ams-cla): Excellent.
Baluty (pl-waw-rodc2): don't understand what levels are what's identities and I also noticed that some of the customers using some specific deployment tools and these tools tools as I understand they are adding these tools adding some levels.

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Baluty (pl-waw-rodc2): So even though for example like you you see that they have one or two policies and they target few levels but they somehow exhausted already uh 64k um and yeah I think it's one of the big problem and we need to somehow address it because at least in Google point of view like Google wants to support lot of lot of pots lot of lot of nodes and um somehow is not great to tell the people that oh you can have 3k 300k pods and you can have only 64k unique levels. Um uh yeah and second one is that so um I think we can go to the next slide and I I
Hoge Veluwe (nl-ams-cla): Okay.
Baluty (pl-waw-rodc2): can talk
Hoge Veluwe (nl-ams-cla): Can I have a question? Uh so does does it not help if you tell your users guess you need to like limit what uh what the relevant labels are I think at some level like because we had to do that so so for for users who have a decent understanding engagement with selium itself that can work right but I think there's a large number of users out there maybe selium just come platform or you know whatever else and I think just saying, "Hey, here's this obscure detail deep in the thing.

00:06:57

Hoge Veluwe (nl-ams-cla): Go and like find like look up and configure this thing way is like not a very user friendly. doesn't want to use labels. So the question is like how do you improve the experience in my mind it's like do we need to push that into upstream as a concept that's as well like there's the whole discussion about like should service accounts bey the other one is like you have that
Baluty (pl-waw-rodc2): All
Hoge Veluwe (nl-ams-cla): and then you have just the blindly blunder into it and then you have the people who are like these
Baluty (pl-waw-rodc2): right.
Hoge Veluwe (nl-ams-cla): central ops teams like try to get their users up with the right thing but they don't full fully have So I think we've seen like all the range. It's just like well yeah my user do crazy stuff and I can't like necessarily like force them. So it's right now it's like yes probably there is a solution with the security labels filtering but it's just not very well known and sometimes people don't have control to it like

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Baluty (pl-waw-rodc2): Yeah, that's
Hoge Veluwe (nl-ams-cla): in terms of the the design how to be optimal about sharing this information the idea is like any
Baluty (pl-waw-rodc2): it.
Hoge Veluwe (nl-ams-cla): given security identity should be some box around some set of workloads that has a different posture respect to any other. Yeah. And if you can make those groupings relatively large then which is probably more common like let's say you deploy 10 well I guess like how many deployments do you have maybe the question like that would be a super micros micro zero trust yeah kind of approach to say each individual deployment has a different security posture everything else but that's a relatively complicated way to do it much simpler would be like if tie directly service account name space right and then it's the question how many of those would you have and then from a silly perspective how do we tie those together with the user experience yes like if we were opinionated and we just said it has to be a name space then I think we would eliminate a lot of these yes um maybe I don't know if it's all but like yeah so I do think um putting my kubernetes hat on it's like yeah have better emotions It just seems like right now the the fact that any label can kind of get

00:09:19

Hoge Veluwe (nl-ams-cla): into relevance like it's kind of baked into the neighborh on that depends on how many keys. Yes. And and labels are used for way too many purposes on there. which is fine like it's just not right. Um, do you want me to go to the next slide?
Baluty (pl-waw-rodc2): Yeah. Yeah. Sure. Yeah. Because I have more details in separate.
Hoge Veluwe (nl-ams-cla): Okay.
Baluty (pl-waw-rodc2): Yeah. Like as I said um this is kind of problem and we see in Google like very lower level and as we want to support very higher scale it will getting more more of and really like as I as I said people I don't know sometimes it helps to give them some manuals what labels are and how what sum identities is and how it use labels uh but as I understand most difficult thing non non like I don't know tech savvy people is that to understand that um yes they using some network policy for some levels but still when they create some pods with random um uh labels it still exhausts the number of identities so and then suddenly oh we cannot schedule a new post

because suddenly it's not able to allocate new identities and also here I I see um so in GKE uh as I understand we don't use cluster mesh and in the code I see that um cluster mesh uh part of this uh identity where is identity can somehow steal the bits from the identity space

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Baluty (pl-waw-rodc2): but not the otherwise why is that are We can we consider the way that if uh someone does not use the cluster mesh, we can steal the B bits from the cluster mesh part.
Hoge Veluwe (nl-ams-cla): Like I I kind of want to like return to the the the kind of previous discussion. Like are you are you are you saying that in a in a single environment you want to express relationships between 65,000 different applications that somehow have different security posture each from one another because like like pods is not the right scale parameter to think about here because
Baluty (pl-waw-rodc2): Yeah,
Hoge Veluwe (nl-ams-cla): like
Baluty (pl-waw-rodc2): again like I I I have seen ton of customer issues uh exhausting the identities.
Hoge Veluwe (nl-ams-cla): yeah I I understand that right but like it necessarily say that uh the solution is to change like how we're expressing this. There's a range of
Baluty (pl-waw-rodc2): Yes,
Hoge Veluwe (nl-ams-cla): possible
Baluty (pl-waw-rodc2): but like what I'm saying is that like saying when that we going to support like I don't know like 400k 500k or something pots and then say that you can have only unique 64k levels.

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Baluty (pl-waw-rodc2): If I were user and if I wanted to have that many parts, I would also think that I can treat them as a separately.
Hoge Veluwe (nl-ams-cla): Do you want to
Baluty (pl-waw-rodc2): I I I mean I doubt there are a lot of companies who wants to have that many uh
Hoge Veluwe (nl-ams-cla): write?

Baluty (pl-waw-rodc2): network policies. Maybe I I don't know. Uh but in theory this is kind of problem.
Hoge Veluwe (nl-ams-cla): So I think there was that um wasn't there proposal to basically only to like dynamically the security relevant labels by looking at the network policy object and then this resulted in a problem right that we discussed I think yeah so yeah because like technically speaking that is the only ones that you should care. Right. But then but then then every time you churn network I I do wonder though is there like some way to slow roll that like let's assume that 60 like 64k plus doesn't make sense right but people have some way smaller set of relevant network policies like is there a way that you somehow you give it like a overseer that figures out I'm in this state right now.

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Hoge Veluwe (nl-ams-cla): I'm going to need this many. How do I sort of migrate things over such that I Yeah. Like imagine you have like a 10x like you just be like I'm willing to to facilitate migrations and all this stuff to consume like 10x space and somehow like move the identities to reduce that churn. I think that that's that's interesting. Yeah. like because it seems like it it's like it it can be dynamic but we know if it's dynamic like super dynamic then you get all these roll out issues like stop working. It's like fine don't go to that extreme go somewhere in the middle but it would conceivably still fit the yeah that's that's definitely an option. I I I think the the dynamic aspect is also interesting because question is like when you're writing these network policies is the kind of posture of those applications like highly dynamic like what is it that's actually changing about how you're trying to lock down the protocol access or the these locations to these locations like is the the fundamental constraint on the network policy there actually dynamic you're trying to do it like I don't know 8 a.m. in the morning 8:00 p.m.

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Hoge Veluwe (nl-ams-cla): and like it always happens in that way or is it like more like oh GDPR compliant you know or like you know you can only access the fraud from the fraud and you know those kinds of things which are more static inherently. Yeah, I think I mean so I mean we can solve the hard you know dynamic problem. I guess I'm saying like if it's if if like is that the problem I guess is like are we expressing the network policy

in highly dynamic? Yeah. So like yeah it'll be it would be an interesting study to look at the network policy to see like because we know the network's policy itself is static and it's really about like adding more policies probably like the rate of change is long term high, right? But then yeah, most of it the same applications or a new version of the application that doesn't shouldn't factor in if this is ideally, right? Yeah. But then what you're saying is like the network policy which right like imagine like the the label set to identity map.

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Hoge Veluwe (nl-ams-cla): You basically compute that whole thing and then you just look at computing the deltas and like presumably the deltas could be very small, right? Like that's that's something that I think would be interesting to look at. I did want to touch on because you asked a specific question earlier about the the usage of the IMG. So you got like a 24-bit mainly focus on 24 bit.
Baluty (pl-waw-rodc2): Yeah. Yeah. Yeah.
Hoge Veluwe (nl-ams-cla): It's a 32-bit within a single node and so we do few extra bits only within the local mode. But generally we try to stick the identity within 24 bits because then it fits within a VXL DNI header transfer it between nodes for that particular deployment model.
Baluty (pl-waw-rodc2): Yeah.
Hoge Veluwe (nl-ams-cla): Then we break that down. We have an option to choose how many bits are actually assigned for cluster ID versus the rest. Um and I think
Baluty (pl-waw-rodc2): But yeah and as I see like if uh someone wants to have in cluster mesh more than I think

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Hoge Veluwe (nl-ams-cla): really
Baluty (pl-waw-rodc2): by default 256 but if they want to have more so it can steal bits from this identity part. So and not the otherwise.
Hoge Veluwe (nl-ams-cla): It's like a trade-off. You can either I think there's only really two drive is like uh two 255
Baluty (pl-waw-rodc2): Yeah.
Hoge Veluwe (nl-ams-cla): clusters and each

Baluty (pl-waw-rodc2): Yeah.
Hoge Veluwe (nl-ams-cla): one.
Baluty (pl-waw-rodc2): And my question is like if someone does not want to use cluster mesh at all, why not let them to use this cluster mesh bits cluster
Hoge Veluwe (nl-ams-cla): I think the concern I would have is actually related to maybe one year ago there's a lot of discussion about people
Baluty (pl-waw-rodc2): bits?
Hoge Veluwe (nl-ams-cla): spinning up different clusters and later on wanting to adopt cluster mesh and then have problems with
Baluty (pl-waw-rodc2): Okay.
Hoge Veluwe (nl-ams-cla): conflicting because like if like if you can lock down that architecture right now forever and never going to want to use that stuff which admittedly maybe talking about um then that's that's one thing but I think in the rest of the soling community I think we find a lot of people main like how do is also but could it be attunable?

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Hoge Veluwe (nl-ams-cla): It sounds like it's not you can't go the other way. I think the other thing is like I just I just think we're trying to solve wrong but like 65k unique apps and unique relationships between
Baluty (pl-waw-rodc2): I mean in theory I can imagine that some customers maybe need that. Uh but again I I don't know like as I said like if I were customer and I need to have I don't know so many pots I would probably think in my mind that I can treat them as a separately. So if I want to have this different workloads that many different workloads I need to somehow address that like but yeah in practice I I doubt maybe a lot of companies need that but
Hoge Veluwe (nl-ams-cla): I mean it's strange that you wouldn't run into like other
Baluty (pl-waw-rodc2): yeah
Hoge Veluwe (nl-ams-cla): problems. I mean if you had that many unique identities now imagine that you each it's not always like single pod right you you have like hundreds of replicas of of the same one. So you would have like 60 4K * 100 let's say or or even 50. It's like huge.

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Hoge Veluwe (nl-ams-cla): It's strange that the the Kubernetes API server would would handle that amount you're talking about network policies object. Well, that's actually I mean part of part part of the work is testing that go that high. But um do you want to talk about the IP
Baluty (pl-waw-rodc2): Yeah. So second part is that I see that it's capped like 500 to 12k and again
Hoge Veluwe (nl-ams-cla): cache?
Baluty (pl-waw-rodc2): like if we get this scenario that like someone has 300k pots and they want to use the dual stack so it's already exhausts the space um and again like what I see that this is the problems we can get in very higher scale but I think Google wants to go that way so we need to somehow rest.
Hoge Veluwe (nl-ams-cla): Is is that just a con or Yeah. Yeah. I think to be honest,
Baluty (pl-waw-rodc2): So
Hoge Veluwe (nl-ams-cla): I'm not sure whether we didn't like should this just be configurable somehow and just you know like find you more memory. I think it already is configurable.

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Hoge Veluwe (nl-ams-cla): I feel like there's some truth to like there is a harder limit there somewhere. Like I'm guess probably and try to change the
Baluty (pl-waw-rodc2): Yeah. Yeah. So my question is more like is it like some hard you know maximum we can get like
Hoge Veluwe (nl-ams-cla): number.
Baluty (pl-waw-rodc2): maybe there is some kernel restrictions or something because I I I don't know
Hoge Veluwe (nl-ams-cla): I feel like this has come up in the past and the limit was harder than I expected it to be, but I don't remember why that's the number. Have you tried it out in search and
Baluty (pl-waw-rodc2): No, I did not. I did not because as I understand you can change and build a new image and
Hoge Veluwe (nl-ams-cla): replace?
Baluty (pl-waw-rodc2): probably it will build up. But my question is like maybe someone knows like it's maybe hard limit or there is I I I assume that when it was designed like this is huge number and we didn't think that like we ever go that's uh that's much high

scale but still um yeah

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Hoge Veluwe (nl-ams-cla): I think whoever has that context is like I would say
Baluty (pl-waw-rodc2): could be
Hoge Veluwe (nl-ams-cla): like like that one feels like potentially a simpler like this just feels like a memory. Yeah. Should we file an issue for that? Sure. Yeah. But then we can maybe file an issue for that and then we can see going on kind of comments on
Baluty (pl-waw-rodc2): Um sure. Yeah. So then we can go next slide then.
Hoge Veluwe (nl-ams-cla): it.
Baluty (pl-waw-rodc2): Uh yeah. So I see that there is uh this mutex um and yeah it works that way and in very high scale yeah it's kind of getting slower and slower. Uh if I did not miss something it works that way. So um I mean
Hoge Veluwe (nl-ams-cla): Do you have a chest that can demonstrate this?
Baluty (pl-waw-rodc2): I cannot like because it's like very high scale but like even very small scale you can notice that so I I made some tests like um because I I my tests done in kind and kind does not you know support the high scale and uh again to run this uh uh this kind of testing very high scale table first of all in Google like we need to get the resource and to get the time for that resource it takes time um but yeah like

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Hoge Veluwe (nl-ams-cla): Sorry.
Baluty (pl-waw-rodc2): it's not hard to imagine how it's done and when you get very high scale you get the issues so yeah this is how it's done um and um yeah
Hoge Veluwe (nl-ams-cla): Okay.
Baluty (pl-waw-rodc2): same with this um regener endpoint regeneration and I think um I think it's noticeable I think other users other than Google too so yeah I'm not saying that what is done is bad but like in very higher scale it's very noticeable
Hoge Veluwe (nl-ams-cla): Yeah, I'd say this is one where we've put a lot of work into optimizing more like P churn for static policy than we have put work into
Baluty (pl-waw-rodc2): Yeah. And what I see, for example, I I I don't know if I can say the customer name,
Hoge Veluwe (nl-ams-cla): your

Baluty (pl-waw-rodc2): but I I I I don't know. Probably not. But what I noticed, yeah, they have really high churn. Like I I don't know. Like I don't know what they're doing exactly, but they killing pots like super quickly and making new pots super quickly.

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Baluty (pl-waw-rodc2): And yeah, they get these problems all the time.
Hoge Veluwe (nl-ams-cla): Right. But like this first one that should not be related at all.
Baluty (pl-waw-rodc2): Yeah. Yeah.
Hoge Veluwe (nl-ams-cla): Right.
Baluty (pl-waw-rodc2): Second one.
Hoge Veluwe (nl-ams-cla): Second one complete. Okay. Yes. Yes.
Baluty (pl-waw-rodc2): Yeah.
Hoge Veluwe (nl-ams-cla): So I think like um the first one I think we probably could demonstrate some sort of benchmark, right? understand how it behaves and wonder like if not
Baluty (pl-waw-rodc2): Yeah, but like it's
Hoge Veluwe (nl-ams-cla): just so yes you could have an end to end test with like 20k nodes but maybe just with the policy stack instead of wiring it up like you could demonstrate just like how fast do you Yeah maybe Thank
Baluty (pl-waw-rodc2): Yeah. Yes. Yes,
Hoge Veluwe (nl-ams-cla): you.
Baluty (pl-waw-rodc2): it could be done. But again like if you check how this um mutx and this thing is used like it used every uh a lot of places. So uh you it's get slower in high

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Hoge Veluwe (nl-ams-cla): Y it's hard to change that
Baluty (pl-waw-rodc2): scale.
Hoge Veluwe (nl-ams-cla): but like I definitely welcome some um like any help I guess in this area right to like learn more yeah where are the I mean I guess the bottom line is like it's it's the global matrix that that's limiting but um so I I think like for me it would be good to have that unit test level benchmark because even if for example we say we propose a fix like how would you know that the fix actually worked well without that test

and then regarding the compilation log is there anything upcoming that would kind of Yes. So this is kind of a a very different one but I think this is more tied into the like clang free kind of stuff. I see. Would clang free fix it. So, so like we have this compil compilation look and that's fine. But like I mean on average what I would expect today is on startups psyllium compiles a template pf program.

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Hoge Veluwe (nl-ams-cla): Yeah. And then you're just taking that binary, you're substituting new parameters and then you pass things down kernel and the kernel has a single global lock for loading, right? And that I think is actually where the majority of the time is spent. So like regardless of what we do in selium, it's like there are things we can optimize maybe. I we already did a lot of optimization compilation but like unless we start to break that down like load smaller programs and re those programs I don't
Baluty (pl-waw-rodc2): Yeah. Is that
Hoge Veluwe (nl-ams-cla): know if aium change I mean the change is to use data infrastructure kernel which is why yeah but s to understand this like the full rebuild cause a compilation because isn't it just a or no. Yeah. So like if you're seeing the compilation block run beyond the first startup, that's interesting. Like everything after that should be parameterized to prevent compilation. That I I wouldn't expect that to be the bottleneck.

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Hoge Veluwe (nl-ams-cla): So if it is, then that's great. Like submit the data, let's figure out why that is. Um, if not, maybe we can
Baluty (pl-waw-rodc2): Um so for example like few customers I definitely noticed that
Hoge Veluwe (nl-ams-cla): look.
Baluty (pl-waw-rodc2): I cannot prove that that was the reason this is the um bottlenecks what I just started taking a look how it works and this is what I see it's complicated and bottleneck at this scale and I think it's definitely like that. And as I
Hoge Veluwe (nl-ams-cla): Like what is the exact do you know or like is it is it like the
Baluty (pl-waw-rodc2): s
Hoge Veluwe (nl-ams-cla): regeneration was it give regeneration
Baluty (pl-waw-rodc2): uh I definitely regeneration time definitely like uh for example I

remember that we have
Hoge Veluwe (nl-ams-cla): time?
Baluty (pl-waw-rodc2): this log I think selum has this log like regeneration all end points or something uh and yeah it takes lot of time for some customers um
Hoge Veluwe (nl-ams-cla): Yeah.
Baluty (pl-waw-rodc2): uh
Hoge Veluwe (nl-ams-cla): So I I could provide a little bit of context there.

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Hoge Veluwe (nl-ams-cla): Um like early on we created this concept regenerate which was more or less recalculate all possible state for any change that happened in the endoint or like you might talk about it in con context of a particular endpoint or pod. Um in which case we'd say you regenerate that and it's any state that needs to change we will calculate that. over the last, you know, eight years or so, we've optimized a whole bunch of that stuff. So, when we talk about regenerate, we kind of say, well, it's whatever is needed, but there are a lot of shortcuts that cut different pieces out. Um and so you may see regeneration time takes a long time but then for instance maybe compilation log like I don't know if we grab that only when compiling or if we grab it while loading the kernel as well like how we actually slice up that overall regeneration time and and measure different portions of that will be really good to know. Yeah, Sunday like I think um given you have seen this in production like maybe we can have a reduction test case to like understand kind of map to that customer use case because then we can figure out you know is it indeed the global compilation log or is it something else that we're generating optimized

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Baluty (pl-waw-rodc2): Yeah. Yeah. Yeah. what I'm saying that like this is bot I see and like and I think it makes sense but yeah the concrete case why some of the customers it took a long time but yeah it needs to be investigated.
Hoge Veluwe (nl-ams-cla): So from what I recall, the metric does break down some parameters, but I don't know if it's as granular as it needs to be for what the bottlenecks are. So we could also potentially extend that metric.
Baluty (pl-waw-rodc2): Yeah.

Hoge Veluwe (nl-ams-cla): Yeah, that sounds like a to just have different stands or something to make sure that
Baluty (pl-waw-rodc2): Yeah. Yeah.
Hoge Veluwe (nl-ams-cla): time
Baluty (pl-waw-rodc2): And the last slide. Uh so I'm not the expert L7 proxy here because in GK uh we don't support NY uh at some point. But uh yeah I see that um this function creator update redirect is called the every filter. Um yeah so can take a lot of CPU time doing that. So but again I'm not the expert.

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Baluty (pl-waw-rodc2): This is what I see and I think in scale it will be really really problematic. Um but
Hoge Veluwe (nl-ams-cla): I feel like there was some optimizations in this recently.
Baluty (pl-waw-rodc2): yeah
Hoge Veluwe (nl-ams-cla): Like I suspect we might not have done incremental XDS updates at all. I don't remember if that's what we were trying to change recently. I don't know. Do you touch this Marco at all or do you hear anything? We kind of for a long time, but we our own and it helped us a lot from updates.
Marco Iorio: Joe,
Hoge Veluwe (nl-ams-cla): So, I imagine should be done as well pretty soon. I didn't.
Marco Iorio: were you asking me?
Hoge Veluwe (nl-ams-cla): Yeah. I was just curious if you had any um
Marco Iorio: Uh, no. Zero context.
Hoge Veluwe (nl-ams-cla): context.
Marco Iorio: I'm not the right Marco for that.
Hoge Veluwe (nl-ams-cla): Yeah. Yeah.
Marco Iorio: Sorry.
Hoge Veluwe (nl-ams-cla): No worries. Okay.
Baluty (pl-waw-rodc2): Uh yeah and the next one I think some people who were on the sick policy meeting I already talked about that.

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Baluty (pl-waw-rodc2): Uh so I had the idea to make um arena shared map um in the

policy engine. Um and this is the way we can achieve the maximum um space saving. Um yeah because of that duplication um and uh uh um the outcome of the meeting on sick policy was that that maybe it's a good idea but because of uh complication of the support of the arena and plus it's only supported 6.9 plus kernel and selium needs to still release few uh releases in with older version of kernels. Maybe we can consider it in the future. Um but now I'm thinking what we can do if we can have the
Hoge Veluwe (nl-ams-cla): Sorry.
Baluty (pl-waw-rodc2): shared not Arena but LPM try uh and I have few ideas um but unfortunately we cannot the achieve the same pure
Hoge Veluwe (nl-ams-cla): Thanks.
Baluty (pl-waw-rodc2): duplication as in Arena uh but still I think uh it will be beneficial
Hoge Veluwe (nl-ams-cla): Cool.
Baluty (pl-waw-rodc2): for a lot of big customers uh and maybe this week or next week of comment meeting. I will share more

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Hoge Veluwe (nl-ams-cla): I think um around arenas like Joe,
Baluty (pl-waw-rodc2): details.
Hoge Veluwe (nl-ams-cla): I don't know what the general sense is on how stable that is, how shaken out. It It's really cool. But also it feels like you can blow yourself up.
Baluty (pl-waw-rodc2): Yeah, I agree with it's super cool. But again,
Hoge Veluwe (nl-ams-cla): Yeah.
Baluty (pl-waw-rodc2): like also information about it, it's like very few you can find, which is weird. But yeah, and even when I was making my PLC, like uh there are some cases it's not documented at all. Otherwise, you have to try and figure out what to do, which is really annoying. Um uh but yeah, idea is super cool. And again this is the way we can achieve the maximum uh space saving we can get in policy engine. Oh
Hoge Veluwe (nl-ams-cla): Kelly, what is the experience
Baluty (pl-waw-rodc2): yeah. Uh it's as I said it's kind of pain in the ass.
Hoge Veluwe (nl-ams-cla): with?
Baluty (pl-waw-rodc2): uh because verifier does not like something for example like I mentioned in this um discussion that so whole idea is having arena that you can use the arena um pointers but same time when I tried to cast the address to arena pointer verifier always throwing some error and I did not make to uh work like to cast the in RNR pointer.

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Baluty (pl-waw-rodc2): So um I don't know maybe I was making some mistake or that's why I still had to use some sys call I I I don't remember what was the name but yeah
Hoge Veluwe (nl-ams-cla): Jesus.
Baluty (pl-waw-rodc2): even with that like when I compare the true etc like it was very pacy mode so um yeah
Hoge Veluwe (nl-ams-cla): I think like anything in BTF like if you have a model about how the kernel proved going to be safe which ideally you get from actually knowing exactly how it works which yes if you wrote the feature that's great but like that that basis of knowledge allows you to do all like whatever it is that you know the feature was designed for that knowledge is probably not documented or well well shared. Um I think was probably introduced a couple of years ago. So there might be a recording from that discussion. Um so that might help for the kind of the basic problems you're facing there. But um one thing that's maybe also worth um raising is I think Teemo had shared um pure use space go arena discussions and and like as I understand like in go there's this proposal to allow memory allocation and like that one I thought was more of like a language issue because as I remember reading through that one it was like they had to basically like mess with every single interface or something.

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Hoge Veluwe (nl-ams-cla): It just became very like this scrolling thing. So I don't remember exactly what the discussion was but there's certainly an element of like who owns this memory and then like what is expecting to clean it up which we are talking about sharing the same go interaction and kernel space. Yes. Yeah. So there's definitely some complications there. Yes. So I think like yeah I think we're probably all in similar like it seems interesting we get this big bulb of memory can do any arbitrary yeah you know structure whatever but my my personal take on it is like you know it probably does make sense for someone to experimental with arena but probably not policy policy is probably not start policy yeah um but if if like we could find some experimental thing to get some knowledge about how to deal with it. That would make sense. I think it's also like um the sort of if we start with like oh can we use arenas for this like it's it's sort of like oh here's a particular implementation we could use but what we should be talking

about is like here's the data format I'm expecting here's like the accesses I'm expecting here's this data structure that I want to express so that I can get either memory efficiency or processing efficiency and then we can like start from that and then say okay well do we have a structure that looks like that and arenas certainly will allow us to build that,

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Hoge Veluwe (nl-ams-cla): right? But then there's this whole other like mountain to climb to to get there. Yeah. Um so then the question is like exactly how far down this do we want to go for the current optimization we need current scale target right so we don't get you know bogged down too much where we can make some progress and in that in that we can also learn more about like we can share the knowledge of the policy agent so in like if there was like a oh here's why LPM try is the wrong way to express policy that would be a really interesting you know discussion right
Baluty (pl-waw-rodc2): and no I I think we cannot avoid uh LPM prime because of this uh port ranges. So um yeah even in my PLC I tried to avoid it but unfortunately I did not find any way to satisfy verifier and not use LPM fry. So I think I saw that it was introduced to support the port ranges and it is the correct way. So I don't think so there is better way we can achieve.

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Hoge Veluwe (nl-ams-cla): Yeah,
Baluty (pl-waw-rodc2): So
Hoge Veluwe (nl-ams-cla): like if if so like do we because like I guess the power of arena is it's a blow of memory. We can choose whatever
Baluty (pl-waw-rodc2): yes, exactly. Yeah.
Hoge Veluwe (nl-ams-cla): way.
Baluty (pl-waw-rodc2): But again like in my PLC like this is the the duplication you can have. So you can have only once um define the rules and you have just reference counts and yeah you just say once and it's quite good in some cases but yeah I totally agree it's really complicated and for example in so I think Timo commented me that I had to use the BPF uh implementation but for example I definitely remember that uh I had to modify this uh go wrapper ebpf some of the places because it's not quite fully supports

the arena especially when I needed to make it the persistent um it did not consider some of the headers or something so I had to modify um evf also
Hoge Veluwe (nl-ams-cla): like the the user space go library right that's fine

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Baluty (pl-waw-rodc2): Yeah, yeah, yeah. If you have library Yeah.
Hoge Veluwe (nl-ams-cla): like I don't know if others are
Baluty (pl-waw-rodc2): Yeah. No, no,
Hoge Veluwe (nl-ams-cla): using
Baluty (pl-waw-rodc2): my point is that like RNA is super new and super complicated and what I'm saying that maybe uh current EVPF uh library does not supports the everything you can do with the uh arena. So yeah.
Hoge Veluwe (nl-ams-cla): I mean that that one seems legitimate. Yeah. Like I think it's just a matter of people actually encountering this. So like Facebook doesn't use because they don't use Go. So basically whoever is using Go who wants to use this feature which is
Baluty (pl-waw-rodc2): Yeah. And as I said like for example like you need to have the fixed address otherwise when you want to uh remap the same memory previously you cannot do otherwise without fixed address. This is kind of like if you want to have RNA you need to have fixed address.
Hoge Veluwe (nl-ams-cla): So I I guess like thinking through like if someone sent a for example S sent a patch to like add better arena support to the go library that that makes sense.

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Hoge Veluwe (nl-ams-cla): I mean I I don't think it's like fundamentally the wrong thing but there's a lot that goes into the implementation details. So like it's completely up to that. Okay. team and and a couple others would be the best position to speak to that probably Dylan I should have asked earlier but can explain how how the arena policy map uh works I mean why I mean I I understand that we have policy map per each endpoint and this will mean that what's common for all of these endpoints can be configured in this shared lucy map. uh can or cannot the or I mean I I understand that we potentially could have like policy map attached to zero right you could share the

policy map without using ar that is technically thing yeah and why is this better or why
Baluty (pl-waw-rodc2): a
Hoge Veluwe (nl-ams-cla): why I think this is like an arbitrary data structure probably the
Baluty (pl-waw-rodc2): big.
Hoge Veluwe (nl-ams-cla): most at least for
Baluty (pl-waw-rodc2): Yes. And like for I mean how can you imagine the current like if you have this LPN how can you make the saving rules only once?

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Hoge Veluwe (nl-ams-cla): So like imagine like instead of using the
Baluty (pl-waw-rodc2): Yeah, what I'm saying that with the arena we can achieve the true the duplication the duplication per rule set and the per rule too.
Hoge Veluwe (nl-ams-cla): I mean that by itself is kind of interesting. suggest to me a memory optimization where typically in the data path we've been more focused on CPU optimization where like if it's you know if we're a little bit slower in updates and we have to do it five times we'd rather do that then have files right because like at some point if you completely duplicate you probably have to have an iteration that allows you to yeah yeah like you don't reach the same you know hashmap but anyways I think the it sounds like let's look at the problem and try to tease out like it feels like there are incremental things that can be done that would kind of get quite a bit I think like two concrete things like one would be for the stuff like named port where it's actually can potentially create point as in security identity is not a defined parameter for it because multiple pods can have the different name.

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Hoge Veluwe (nl-ams-cla): So like how you abstract that away would be one problem statement. Then separately how would you group by security identity. So all pods on the same node with the same security identity same that would be another problem. You couldn't solve that without solving the first problem. Otherwise you'd have the wrong implementation behavior just like those are two big chunks. I think there's still some additional ideas you have there which I haven't covered with those two but like those two by themselves are also over chunk of

Baluty (pl-waw-rodc2): Yeah, sure.
Hoge Veluwe (nl-ams-cla): okay moving on okay so just briefly tracing I don't know if the people so basically um I think tracing went in the CFP but like I don't know it was like a couple months ago or something but I think what was missing was like it basically said if you injected packets like this then it will dump it out. So um there are some folks in our team looking at basically completing that on the other side which is the flowj um I don't think there's anything like shockingly uh unique about this basically it's just like how do you generate the packets but then on the other side um I think in their CFP they'll probably in discussion they go into detail about kind of how they think it could be Um but the feedback on

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Hoge Veluwe (nl-ams-cla): the talk about scalability tests the drive. Yeah. So um I just like um uh yeah so so basically what I want to like uh start discussion about uh like uh how could we uh uh so my team is uh responsible for the the availability in Google. Basically we are interested in like we want to uh hear your feedback and also like uh how how we can uh better supports and and for example can we employ the um like uh more testing the selium in in the as early as possible. Uh uh so yeah so I I'll just start with like u maybe explaining like yeah so one one of the things I think is that um I think like there is there's right now test of like okay let's test like 5,000 nodes whatever 10,000 nodes but those are massively expensive and I don't think like as a develop as a developer you don't want to do this thing where like you have to run a 12-hour test just to figure out your one line change like did something right.

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Hoge Veluwe (nl-ams-cla): So like that's that's something that I think to get to these high level scale we probably need a better approach to get feedback from that without like just like doing literally like like giant thing. Yeah. Yeah. Exactly. So as you can see it's massively expensive we said terrible feedback loop. So yeah, you can see how it looks. The the the working with those tests are extremely hard. Uh and then it's uh it can fail for whatever reason. It can be the infrastructure, it can be one single line of code or or it can be just random failure uh and it's uh then you know you're it reset those hours. Uh so my team is um like uh like this is like probably the main point our main tool and and

main like things that we are working on are scalability test. So we kind of started started experimenting with like idea. Okay. So can we do this better like uh is there is there a way to uh to to improve it and this is kind of my idea.

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Hoge Veluwe (nl-ams-cla): Now this is like very early and uh not well refined idea but this is kind of like to start a discussion and if it's like a good approach or not. So, uh I unfortunately didn't uh like didn't finish my demo. Uh I started working on some you should definitely demo it as a community. Yeah. Yeah. Next time when finally make it work but I built this tool I call it like this work in a cube rewind. So you can get idea what it does. basically uh record the state and all the changes the specified resources. So in this case for example I'm interested in watching notes in endpoints not demo sets and deployment um and then and then I I record all the changes and I can specify like is it specific name space or or or with specific label um it's all configurable then I create a test artifact so I run those very expensive tests like 15k 30k 65k whatever um and then I get this test artifact that in theory I could share with um community and this tool could be used by anyone.

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Hoge Veluwe (nl-ams-cla): So anyone could share uh and we and and um this would allow to reconstruct some state um in in much smaller scale they say kind so I can deploy 15k fake nodes and all the like summing nodes end points from the specific test case like and like okay so why why not like create um so maybe you might have a question like why not create like a shell script where I just trade between pelium nodes um and and uh other resources like that. So what I experience when because I already tried it is that the selium and kubernetes are very complicated, very complex and the dependencies between those resources and in the agent and operator when it processes it they are not that obvious like you can create 15k nodes and it's going to process them in like 30 seconds maybe very quickly but when there is like a a very very specific conditions and we for example it can take 8 minutes right so it's not that simple to just write a test case uh and come up with those model so that's kind of uh where where I come from like okay either you need to know really well the agent uh or operator or selium and maybe also kubernetes and then you are able to like write those tests or

maybe we could come up with some tooling some um some some um uh better tools for the developers that they could I don't know like for example get the um the state capture the state uh either either like a like final state or maybe the sequence

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Hoge Veluwe (nl-ams-cla): that you know came to the state uh and then we can replay the the sequence of events or we could you know just apply those final resources and uh the the idea is that because we found many issues that are very easily reproducible in our test like we constantly run those large scale test and we and we constantly were hitting those problems. Uh so it like so if we could like share with community with all of you like okay so we hit this issue this is how we reproduce and we don't require from you like okay so now scale 50k and test it uh so so this is kind of like um like my idea that I had u when working with this test that I really wish to have like first of all a simpler way to like write this test because it's also not that easy like you work with CL2 know uh like scale it's not that easy to write test case and then like configure everything so it works and and the other uh is that uh also like a framework like I don't know I can write a simple go with with for example those test and run um so so this is kind of the the rough idea of of uh what I'm working on, what I'm experimenting with.

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Hoge Veluwe (nl-ams-cla): They're also interested like in your feedback. I know that there are some synthetic tests, some benchmarks already in sium um and and uh they are fine but again as I said you really need to know you need to understand the failure point like what is causing it uh and to write those tests and and maintain them. Um, so yeah, I'm I'm just curious like what are your thoughts on it? Does it make sense? Does it make sense? Am I right that there is some uh nothing in that text? So we pretty much the similar idea we are faking uh all the state into loading into CDL that state the same is there I do remember myself doing exactly that style like two nodes and then like how many in my opinion on this highly valuable have that I think the stuff where you know like you don't have Google trust Google I mean like one thing do you because technically you could also just run but I guess you want really sure that it's the it's hard to precon of all the interactions and I think that's that's one of the difficult that's why like people go to the actual test it's like oh I ran it at 15k for sure I could even imagine like maybe you're a

bit more manual and doing all sorts of interesting things just as a more like a

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Hoge Veluwe (nl-ams-cla): very user kind of level on this side and you don't necessarily know that effect it's going to have record what you did then find like oh I experienced this weird condition when I did added these objects and these objects But like in the reconstruction you so you fake every not but you just want test on what the behavior okay yeah yeah so so let's say we have for example GK right and the control plane okay then you can just operate like that so uh you need to use something like which is basic which is kubernetes without cublet um and it's basically like you marks the have checks for the nodes for the ports and stuff like this. So it doesn't delete for example when you just have to see notes uh after some point operator garbage collect them. So um we need something like that. Uh uh so so the idea is to like not like in my mind like perfect uh scenario is that okay I as a user can use this tool to like record the you know if something is reproducible uh constantly and it's not like the race condition something but yeah the race condition usually not that simple uh and I don't think this would help with it but uh so like beyond to and you just simulate the rest of the scenes to Yes.

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Hoge Veluwe (nl-ams-cla): Yes. So, so the qu like uh so I would have the senium fake faults in those fake nodes. Yeah. they just don't really exist just in the API but yeah the idea is that okay I two nodes or one note or 10 nodes and I just you know for example restart those nodes and see if I can see for example the engine start uh issues the same that I had without for example by and for example I also let me make like the uh cube API um uh like overloading uh right So okay with two nodes cube API is probably uh has enough capacity to like support them. Um so I just uh don't care about anything and then also the setup is a bit easier like uh I don't uh like when running those tests I just need the kind you know it's quite easy to set up the kind cluster. Uh so then yeah I can I can uh just you know add the additional command to load this artifacts uh and and have the so so this actually like applies those artifacts to the cube API.

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Hoge Veluwe (nl-ams-cla): Yeah exactly. So the way I designed this tool is that first so there is still like an operator you can deploy operator to the cluster or you know infrastructure that watches the cube API and all the changes for the chosen resources. it saves it in like uh in my case like Google store but it can be locally or you know S3 AWS or somewhere then you can get them like compile those uh resources and you can like specify okay uh for example I am interested to because okay you you have metrics but metrics don't really show the whole story it's especially at scale it's like uh it it it gives you maybe the idea what's going on But you don't really know what's going on. Uh and then you can look at the logs. There's like million of logs and it's impossible. Uh like really hard. So uh what I want with this tool is okay I see some logs that there was a failure. So I can use this tool and use it works like cubectl.

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Hoge Veluwe (nl-ams-cla): It just adds like the like at time. Uh so I can qctl get this resource at design and I see exactly what was uh happening there and also I can see like all the changes for the like how frequently change something like that. So this is like my idea for this tool. Uh it's very simple like nothing complicated. Uh and this is kind of my my uh my goal like I I want to um create a tools that are very easy to use that are not complicated for scale test because I think current tooling they are extremely complex and very hard and working with them for anyone is is uh not a nice experience and I would like try try to bring a bit better uh I don't know how it works. Can you tell how uh from your experience from your test uh how big should those what your flavor usually is when you try to apply like 16 or uh objects to the API? If you're doing it in Kap.

01:01:56

Hoge Veluwe (nl-ams-cla): Yeah, but I was testing with GKE and the the cool things is that uh what you can do is let's say you can we can record those uh 15k tests and then we can run in parallel a couple tests with different uh instance. So we can see for example okay does the increase in the instance and I don't need to run 15k test to test it again. So it's kind of like yeah so so we can't this is another problem we can't create like a matrix for 15k test for different instance like impossible but maybe with it we could so

we can tell okay selium works at this scale with this instance file um and and or agents work so so this is another benefit of it um yeah it should be very yeah but this be even cheaper than our current node maybe I think that's what I found in GitHub work but this is this works well why is the demo right like if it's if it's reflective of the the real environment then I feel like scale we do today scale is also like many different right is is another Another thing that we are trying to figure out um the the number of nodes, number of votes like you get.

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Hoge Veluwe (nl-ams-cla): Yeah. Like running those exactly and then changes in those things, right? The ch all of those resources. It's like if we could like record those those things one like okay we need to run still but then maybe we don't need to run them that often and then we can like I don't know every like uh couple versions every month and even like even if it takes a bit longer if you just say this is the the kind of playback and you run that in your environment you say how long does it take to reconcile or to get into a good state right then like just that length of Time is now a metric. Say, oh, there was a regression at this time because right like that test that used to take 30 minutes now takes 50 minutes. Uh so uh for now what I'm doing is so um I I we we have a couple issues couple scenarios that uh we con that are in the test we were uh always reproducible like for example senior nodes new nodes in DK uh so we constantly hit like the startup uh SLOs's uh limits um and there is like for example also with with s point sizes uh uh so what I'm doing now I'm trying I'm testing my

01:04:54

Hoge Veluwe (nl-ams-cla): tool so to see if I see the same like issues in this uh fake environments uh so once I'm figure out if this tool actually helps with it and and it actually uh like can reproduce those issues because for example we've seen nodes. Um we we found the issue and we actually like have a fix for it uh already open. Um that processing ceiling nodes was uh very time consuming on the startup and it it even took like 8 minutes to start which uh was not enough and the the uh the senium agent was killed before it could finish it. So uh we that one no this one. Uh so yeah I was helping him uh debugging it but debugging it was uh like really really timeconuming very hard because we basically had to add logs uh deploy everything to the DE and then run those

test they fail again uh or or they pass but we have not enough logs for it so we need to add deeper logs. So this is kind of the the um where where this idea comes from.

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Hoge Veluwe (nl-ams-cla): We don't really want to uh like yeah and run selium in in debug mode with 15k nodes. Yeah, it doesn't work. You can't do this. Uh and 5k you can't do this. It just doesn't work. Uh so yeah could be also quite useful for crystal. I think you just have to make two crystal at the same time or something. And uh yeah. Yeah, I think generally this is I've been pushing to like basically get these reductions in scale tests because just like running it at the giant speed feels like we're never going to end like you have to fight with AI. Yes. Um yeah, but but okay. Yeah, I'll try to like uh come up with uh Yeah. So yeah, the goals, right? I already explained it. Uh so so what I think the the next months uh like this this is kind of like my side project uh for now but we'll see how it goes.

01:07:12

Hoge Veluwe (nl-ams-cla): Uh but I guess like what I am particularly interested in uh like how can I you know uh support the the open source sidelium. Uh this is one of my ideas but maybe there are others uh like how how can we like make those tests more maintainable. uh feedback loops faster because I think what one of the problems is you know the those problems like with scalability they often are not that hard to fix it's just a matter of like finding them debugging them like okay what's actually causing those bottlenecks and then uh verifying the the the right so in oss it's for our case it's usually not possible and then regression I think the other one is big is regression it's like very hard to figure oh would be aggressive. I think it's like very very common that this ends up being like this one person whose job it is to figure out all this stuff. Yeah. It'd be nice if we can democratize it a little bit more and say like like if you know I'm writing a new feature I'm like I I think there might be a scale implication.

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Hoge Veluwe (nl-ams-cla): Yes. I don't really want to be the person to introduce the

regression right but if I can get data like this this is nice. Yeah. Yeah. We could try CI/CD probably some of the scenarios which is I think huge cut catching it the the regressions like before PR merge because right now it looks like it's already like very very late uh in the process like usually when we uh already have the deployed the the version GK version then we test it as probably others as well. Uh uh oh yeah that like huge for us and cool and like one comment that if like a user come up with an issue and say hey I see CM agent too much memory or whatever could you just like somehow integrate that into system and say and see what happens is that safe It gets a little more complicated for sure. Yeah. Yeah. It's probably if if we going to like uh share those information for you, we have to add some logic for uh you know changing the service names and stuff like anonymization.

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Hoge Veluwe (nl-ams-cla): Sorry. Anonymization. Yeah. Yeah. Like where we're at right now with the certain open source project is like we provide a tool that will allow you to capture a dump of the current snapshot of the system plus maybe like a couple of minutes of flows. But then if you want to give that to somebody that's 100% your responsibility to figure out what you're comfortable sharing, right? Um, and like I haven't loved the idea of trying to implement something better in the open source because like if we tell you just run this extra flag that will hide all the important information and it doesn't do enough hiding for you then like you're going to Yeah. It puts us in a weird position because it's like it should be your responsibility to figure out what you're comfortable sharing, right? I mean, I guess for now it's just like uh probably going to be like just just for the testing uh purposes and just for like if the tool is available for anybody who can record they can download that and run them in their own that would be kind of neat.

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Hoge Veluwe (nl-ams-cla): I'm not saying it's required like obviously just the uh the reading and results and then replaying is good but if you can record some people might want to record I don't know who exactly yet maybe they should I mean it's just like the whole like if we run it on do you think there would be security implication other than an amazing data like can it run there and I mean find this malware and I run it and I still put

there um I don't I don't know like probably you'd have to finish the yeah so I tried like to for example see like make it as minimal as possible so I made a time chart that uh you specify like which uh resources want to watch and it only make like a rein for this. Yeah. So, uh, be safe. Uh, sorry, we can we can document what it is and like people. It's probably not something you want to run on production just, you know, capture like um I don't know maybe in the future, but it's probably way before uh you're going to like just run it like many optimization be and so on for that.

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Hoge Veluwe (nl-ams-cla): Uh not yet basically it's like the you have you have like database with like log base right well I'm just asking silly question the base thing is already super useful I think cool I mean you got people so excited they want they want to have the fifth version that's it um well thank you foring so um next That's probably demo. Super excited for this. And then Joe, I think you have Is there some link for this? It lives in my laptop for now. Uh yeah, probably will take some time before we or if we open source it. But um yeah, I'll do demo. I'll show like uh the the issues uh we managed to reproduce with it or we if we did not uh maybe it's going to be like there's like it's useless uh this approach or something but then I also would mention that I think it's made some usage and even just like the general like when you explain that it's good replay yeah it would be good to know like if it doesn't work like Why?

01:13:29

Hoge Veluwe (nl-ams-cla): Um, future reference. All right. Did you get Yeah. Okay. I don't know. Still worthwhile to dig through beta pages, but let's just do it. Sure. You just ignore it. Um, so if you're going to leave, just go down to the Okay. I mean, I can walk you out. Don't go up and hide somewhere. I'll walk you out later. All right. Okay, so I um gripped your beta in the trees the other day. Um here's everything that turned up. Um so so broadly I guess as a project we should probably think about what we want to spend our time and effort on making better and what kind of want to retire and I think particularly historically we've tried a bunch of things maybe added that added those things into the code and sometimes worked out really well users adopted it we got great feedback we managed to productionize the features sometimes the features we added and um for whatever reason sort of either nobody used it or you know we just never got on the path

to productionizing it.

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Hoge Veluwe (nl-ams-cla): So from the perspective of just like how do we reduce the maintenance burden for the project it's like worth kind of reviewing. Um, so I kind of just thought of a few categories here like well first like okay yeah what's the feature name like which area of the codebase is it kind of focused on is there like a particular goal in the project overall that we're trying to achieve with that feature. Um how well do we think it's used in the community and then like what kind of making burden do we feel it's actually imposing on us like if it's I have to say 79 point slice is used to count all of GKE in Azure. Yeah. Yeah. Yeah. I think initially I was like I might put some significantly. Yeah. Cuz I I think I think Azure is also using I know all of GKE. I figured like anything that like for the clouds for the scale and stuff is Yeah.

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Hoge Veluwe (nl-ams-cla): Yeah, like that one I think is very significant. So I think it's been a recurring discussion in the um six scalability for years now like was there just one thing left we want to fix before we mark it as not beta but I haven't followed super yeah like maybe that is useful for sig scale we just finally nailed down because I I know that there were some ns and then I was commenting on some stuff but it was just like it truly like you don't need to fix it like I think one of the concerns we had is is um like if you have two features which are both like GA and they don't work together. Oh, it's kind of a a weird I think. Yeah. Then maybe it was like egress gateway something. But there is a PR like Okay. Was it egress? Yes. So yeah. So, some of those like I and I don't know if I've got old categories here and but that one seems like it's just worth pushing, right?

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Hoge Veluwe (nl-ams-cla): Maybe that's the category. Yeah, we we should label it's just like, hey, seems like do it soon. Yeah. Um, so yeah, I I don't know if the folks in this room necessarily can speak to most of these. Um but if there is an item in this list that you want to discuss. Yeah. Uh discuss that. Why does the BPFT protein does not work? It's

cuz I ran out of time on the future three years ago. Um um like where that got to was um yeah I mean like effectively I was kind of ran out of time to work on it but like there was we we came up with permutations testing for different data paths through the proxy through the through the media through the proxy. Um this was like right before this we created the selium CLI CLI connectivity tests. We had some like deployment of just like a YAML thing that would deploy a bunch of pods that talk to one another in different ways.

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Hoge Veluwe (nl-ams-cla): Configure a bunch of network policies that go through different paths. There's one or two of those that didn't always work depending on which node they were on. And maybe I just remember EKS having some weird behavior that failed. So it just didn't end up chasing down all those bugs. Um and there hasn't been that much user demand. Yeah. What do you use Troxy for? All right. Uh if we want to have like the working on nodes classic uh service mesh. Yeah. Okay. So anything involving getting packets into envoy for whether service mesh gateway API stuff or network policy at L7 all that goes through a version of G proxy. This one is taking it out of IP tables and putting it into Oh, I see. Okay. So it's like slightly different implementation. We just haven't chased down. I think one of the latest ones was like Netkit for some reason with Netkit.

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Hoge Veluwe (nl-ams-cla): It's incompatible. Um there's some kernel patch needs to be done. Alistister from our team was digging into that for a while, but I think you hit a bit of a wall. Yeah, I mean it's difficult to to understand what's actually happening. It just, you know, the the record just times out and I don't know why, you know. Yeah, I tried uh pocket where are you? You tried Yeah. TP up, but it just like gets lost somewhere, you know. At one point I did have some idea about like a tasting tool we could try to create to like validate some portion of that. So I think what we need is better tools to be able to reduce the thes like I mean the net filter free project and altogether is like sometimes it would help so much if uh packet where you was possible to use on as far as I know it only works. Oh, I see. Yeah.

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Hoge Veluwe (nl-ams-cla): Yeah. Cuz like let's say you want to use you want to go filter by this IP address. It's probably like filter or something. So like at ACP there's no like the current hasn't extracted that only takes an interesting question. It's an interesting point, but I think for that one also like it's it's kind of an extra relatively small code path. Like I don't think it's actually causing much at the moment. So we just not worth out per se, but so I think um probably uh it would be useful to go over this in the community. Yeah, probably get by. I mean like there's only like a few people here But like I I also scroll down probably the ones that are can there be can this be on community meeting on Wednesday uh let's say 10 a.m. CST. So central European time I guess not. Yeah. I mean the majority of people who work on Selium are probably either US somewhere Europe and quite a few people on the west coast.

01:22:37

Hoge Veluwe (nl-ams-cla): That usually means kind of the 5 p.m. I think maybe at the moment the time zone has shifted such that it's 400 pm. Was it 4 p.m. for you or is it uh It's normally 4 p.m. but earlier. So like next week it will be at 400 p.m. on Wednesday. I think that unless that's when starts new year. But like that's about to stop, right? In fact, it might even be this. Yeah. I mean, we'll be good here. But I mean, if you have input on any particular ones of these then like um I'd be taking some know as well. Yeah. I'm curious about that that kit. Are you guys using that on experimental? Yeah, I'm assuming it's going to it's not getting booted, right? Like a lot of reasons why it's a it's a good future. I think that's there's a there's a column there I'm missing somewhere about but maybe it's partly like the project goal.

01:23:52

Hoge Veluwe (nl-ams-cla): Like I mean I see it as mainly a performance thing and I think it does a bunch of cool new stuff. I think it's very much align project right so I think netkit by itself has a bunch of adoption in big companies but not necessarily in context of yeah I think maybe one or two are but yeah I probably I don't know when we're going

to flip it on I'm just going to assume after some maturity level maybe it's like to do with some support yeah so So maybe arguably the thing the question for this is like how soon and what is our actual like right I mean we want to make a G first and then decide whether I don't know if there's not great considerations yeah I don't if you had to try um I mean I did not only like manually tried to run that worked on its but uh we have not yet I know there was like big hype around that last Yeah, I guess MCS we should probably pull out at some point and see. Well, maybe I imagine that's something we also want to just graduate at some point. There might be some changes coming for MCS. It's actually a keep. Well, actually, it's a good point. Yeah, maybe it's partly subject. Yeah. So, anyways, just I would just probably watch the Okay, I think um thank you everyone who's still here for coming. So, always uh always happy to help post.
Marco Iorio: Thanks.

Transcription ended after 01:26:33

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